

OpenStax Calculus Volume 3

Section 4.8 Exercises: Lagrange Multipliers

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For the following exercises, use the method of Lagrange multipliers to find the maximum and minimum values of the function subject to the given constraints.

1. $f(x, y) = x^2y$, $x^2 + 2y^2 = 6$
2. $f(x, y, z) = xyz$, $x^2 + 2y^2 + 3z^2 = 6$
3. $f(x, y) = xy$, $4x^2 + 8y^2 = 16$
4. $f(x, y) = 4x^3 + y^2$, $2x^2 + y^2 = 1$
5. $f(x, y, z) = x^2 + y^2 + z^2$, $x^4 + y^4 + z^4 = 1$
6. $f(x, y, z) = yz + xy$, $xy = 1$, $y^2 + z^2 = 1$
7. $f(x, y) = x^2 + y^2$, $(x - 1)^2 + 4y^2 = 4$
8. $f(x, y) = 4xy$, $\frac{x^2}{9} + \frac{y^2}{16} = 1$
9. $f(x, y, z) = x + y + z$, $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1$
10. $f(x, y, z) = x + 3y - z$, $x^2 + y^2 + z^2 = 4$
11. $f(x, y, z) = x^2 + y^2 + z^2$, $xyz = 4$
12. Minimize $f(x, y) = x^2 + y^2$ on the hyperbola $xy = 1$.
13. Minimize $f(x, y) = xy$ on the ellipse $b^2x^2 + a^2y^2 = a^2b^2$.
14. Maximize $f(x, y, z) = 2x + 3y + 5z$ on the sphere $x^2 + y^2 + z^2 = 19$.
15. Maximize $f(x, y) = x^2 - y^2$, $x > 0$, $y > 0$;
 $g(x, y) = y - x^2 = 0$
16. The curve $x^3 - y^3 = 1$ is asymptotic to the line $y = x$. Find the point(s) on the curve $x^3 - y^3 = 1$ farthest from the line $y = x$.
17. Maximize $U(x, y) = 8x^{4/5}y^{1/5}$, $4x + 2y = 12$
18. Minimize $f(x, y) = x^2 + y^2$, $x + 2y - 5 = 0$.
19. Maximize $f(x, y) = \sqrt{6 - x^2 - y^2}$, $x + y - 2 = 0$.
20. Minimize $f(x, y, z) = x^2 + y^2 + z^2$, $x + y + z = 1$.

21. Minimize $f(x, y) = x^2 - y^2$ subject to the constraint $x - 2y + 6 = 0$.
22. Minimize $f(x, y, z) = x^2 + y^2 + z^2$ when $x + y + z = 9$ and $x + 2y + 3z = 20$.

For the next group of exercises, use the method of Lagrange multipliers to solve the following applied problems.

23. A pentagon is formed by placing an isosceles triangle on a rectangle, as shown in the diagram. If the perimeter of the pentagon is 10 in., find the lengths of the sides of the pentagon that will maximize the area of the pentagon.
[Figure: A rectangle with an isosceles triangle on top. The side of the isosceles triangle with the two equal angles of size θ overlaps the top length of the rectangle.]
24. A rectangular box without a top (a topless box) is to be made from 12 ft^2 of cardboard. Find the maximum volume of such a box.
25. Find the minimum and maximum distances between the ellipse $x^2 + xy + 2y^2 = 1$ and the origin.
26. Find the point on the surface $x^2 - 2xy + y^2 - x + y = 0$ closest to the point $(1, 2, -3)$.
27. Show that, of all the triangles inscribed in a circle of radius R (see diagram), the equilateral triangle has the largest perimeter.
[Figure: A circle with an equilateral triangle drawn inside of it such that each vertex of the triangle touches the circle.]
28. Find the minimum distance from point $(0, 1)$ to the parabola $x^2 = 4y$.
29. Find the minimum distance from the parabola $y = x^2$ to point $(0, 3)$.
30. Find the minimum distance from the plane $x + y + z = 1$ to point $(2, 1, 1)$.
31. A large container in the shape of a rectangular solid must have a volume of 480 m^3 . The bottom of the container costs $\$5/\text{m}^2$ to construct whereas the top and sides cost $\$3/\text{m}^2$ to construct. Use Lagrange multipliers to find the dimensions of the container of this size that has the minimum cost.
32. Find the point on the line $y = 2x + 3$ that is closest to point $(4, 2)$.
33. Find the point on the plane $4x + 3y + z = 2$ that is closest to the point $(1, -1, 1)$.
34. Find the maximum value of $f(x, y) = \sin x \sin y$, where x and y denote the acute angles of a right triangle. Draw the contours of the function using a CAS.
35. A rectangular solid is contained within a tetrahedron with vertices at $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$, and the origin. The base of the box has dimensions x , y , and the height of the box is z . If the sum of x , y , and z is 1.0, find the dimensions that maximizes the volume of the rectangular solid.
36. [T] By investing x units of labor and y units of capital, a watch manufacturer can produce $P(x, y) = 50x^{0.4}y^{0.6}$ watches. Find the maximum number of watches that can be produced on a budget of $\$20,000$ if labor costs $\$100/\text{unit}$ and capital costs $\$200/\text{unit}$. Use a CAS to sketch a contour plot of the function.